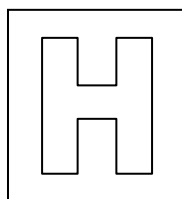


Candidate Name: _____

Class Adm No

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2011 Promotional Examination II Pre-university 2

H1 CHEMISTRY

8872 / 01

Paper 1

22 September 2011

50 min

Additional Materials: OMR
 Data Booklet

READ THESE INSTRUCTIONS FIRST

INSTRUCTIONS TO CANDIDATES

1. Do not turn over this question paper until you are told to do so.
2. Write your name, class and admission number in the spaces provided at the top of this page and on the OMR provided.
3. Answer **ALL** questions and shade the correct answers on the OMR provided using a soft pencil.
4. **No extra time** will be given for shading.
5. Hand in the question paper and the OMR separately.

INFORMATION FOR CANDIDATES

Marks will not be deducted for wrong answers; your total score will be the number of correct answers given.

FOR EXAMINER'S USE	
TOTAL (30 marks)	

This question paper consists of 12 printed pages.

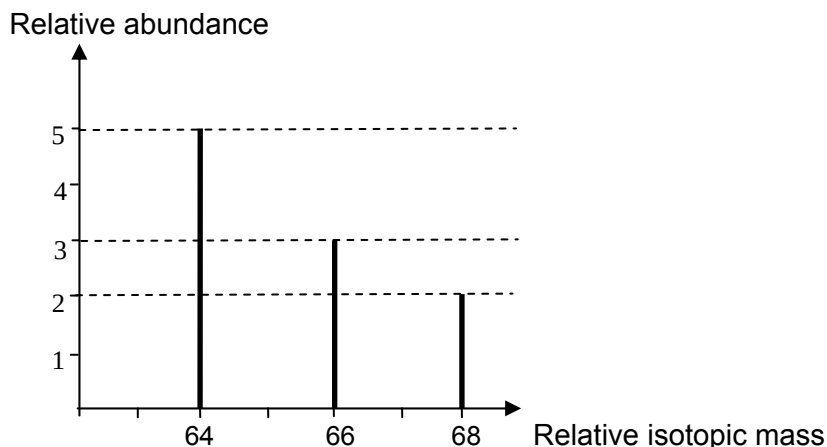
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Answer all questions on the OMR form provided (30 Marks)

For each question, there are four possible answers, **A**, **B**, **C** and **D**.

Choose the **one** you consider correct.

- 1** The relative abundances of naturally-occurring isotopes of a sample of zinc are shown below.



What is the relative atomic mass of zinc in this sample?

- | | |
|-------------|-------------|
| A 63 | C 65 |
| B 64 | D 66 |
- 2** When sodium azide, NaN_3 , is heated to 300°C , it decomposes into its elements rapidly. What is the volume of gas produced when one mol of sodium azide decomposes?

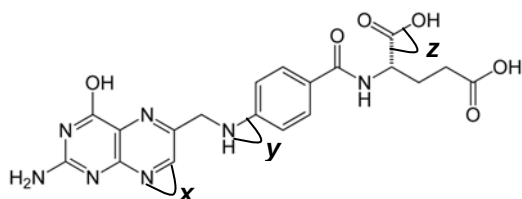
(All volumes are measured at room temperature and pressure.)

- | | |
|----------------------------|----------------------------|
| A 36 dm^3 | C 72 dm^3 |
| B 24 dm^3 | D 48 dm^3 |
- 3** Sulfur dioxide, SO_2 , is added to wines to prevent oxidation of ethanol by air. To determine the amount of SO_2 , a sample of wine was titrated with iodine, I_2 . In this reaction, one mole of SO_2 was oxidised by one mole of I_2 and two moles of electrons were exchanged in the system.

What is the change in oxidation number of sulfur in this reaction?

- | |
|-------------------|
| A +2 to +5 |
| B +2 to +6 |
| C +4 to +2 |
| D +4 to +6 |

- 4 The following is the structure of folic acid.



What are the bond angles marked **x**, **y** and **z**?

	x	y	z
A	180°	120°	109.5°
B	119°	107°	120°
C	120°	120°	120°
D	180°	180°	109.5°

- 5 Ammonium carbonate is a commercial salt with the formula $(\text{NH}_4)_2\text{CO}_3$. It is used when crushed as a smelling salt. It is crushed when needed to revive someone who has fainted. When dissolved in water, NH_4^+ and CO_3^{2-} ions are formed.

What are the shapes of the NH_4^+ and CO_3^{2-} ions?

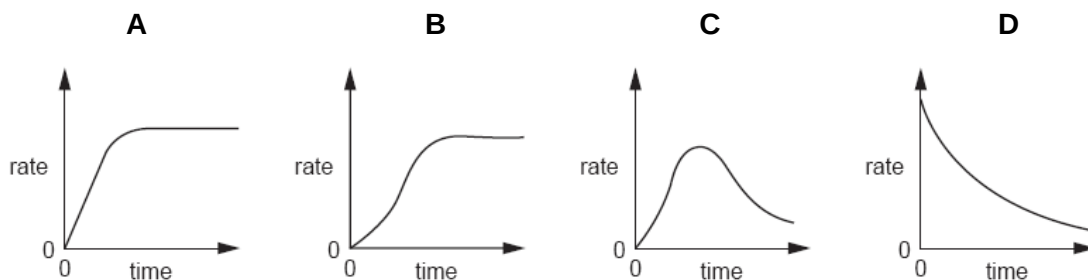
	NH_4^+	CO_3^{2-}
A	Square planar	Trigonal planar
B	Tetrahedral	Trigonal pyramidal
C	Tetrahedral	Trigonal planar
D	Square planar	Trigonal pyramidal

- 6 Which of the following metals is expected to form ionic compounds with the greatest covalent character?

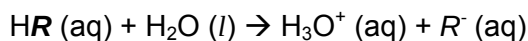
A Barium	C Magnesium
B Calcium	D Potassium

D - 3342

- 10 Which curve is obtained if the rate of reaction is plotted against time for an autocatalytic reaction (i.e. a reaction in which one of the products catalyses the reaction)?



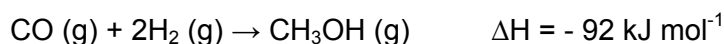
- 11 The hydrolysis of a substance, HR , is first order with respect to HR .



When a 0.1 mol dm^{-3} solution of HR in water is prepared and kept at constant temperature, the concentration of $H_3O^+(aq)$ in the solution after 30 minutes is $0.075 \text{ mol dm}^{-3}$. Which one of the following is the half-life (in minutes) of the reaction?

- A 7.5
 - B 15
 - C 20
 - D 60
- 12 $K_w = 5 \times 10^{-15} \text{ mol}^2 \text{ dm}^{-6}$ at 289 K
 $K_w = 1 \times 10^{-14} \text{ mol}^2 \text{ dm}^{-6}$ at 298 K
 Which of the following can be used to explain this difference in K_w values?
- A $H^+(aq)$ and $OH^-(aq)$ react together exothermically.
 - B The pH of water is independent of the temperature.
 - C The concentration of water is lower at the higher temperature.
 - D K_w is dependent on the concentration of H^+ in water.

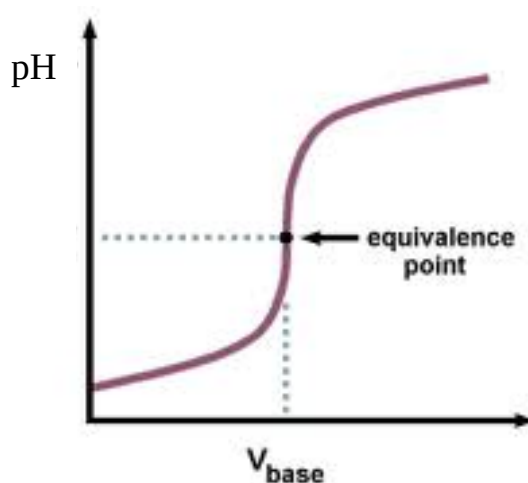
- 13 The manufacture of methanol as shown in the equation below requires conditions of 250°C, 50-100 atm and a catalyst containing a copper compound.



Which statement explains why such conditions are chosen?

- A The rate of formation of methanol increases at low pressure.
 - B The addition of a catalyst increases the equilibrium yield of methanol.
 - C The rate of formation of methanol increases at low temperatures.
 - D The yield of methanol increases at low temperatures.
- 14 The graph below shows a pH curve of a titration between HCl (aq) and NH₃ (aq). The pH of the solution when varying volumes of NH₃ was added was measured using a pH meter and plotted on a graph.

What is the pH of the equivalence point and a suitable indicator to be used in this titration?



	pH at equivalence point	Indicator
A	>7	methyl orange
B	<7	methyl orange
C	<7	phenolphthalein
D	>7	phenolphthalein

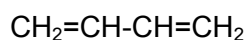
- 15 For the equilibrium of $\text{CaCO}_3(\text{s}) \rightleftharpoons \text{CaO}(\text{s}) + \text{CO}_2(\text{g})$, the equilibrium mixture at 300K contained 1.50 mol of CaCO_3 , 1.20 mol of CaO and 0.380 mol of CO_2 . The reaction was carried out in a 1 dm^3 vessel. Calculate K_c , stating its units.

	K_c	Units
A	0.456	$\text{mol}^2 \text{ dm}^{-6}$
B	3.29	$\text{mol}^{-1} \text{ dm}^3$
C	0.304	mol dm^{-3}
D	0.380	mol dm^{-3}

- 16 Which one of the following compounds **does not** exist in both *cis* and *trans* forms?

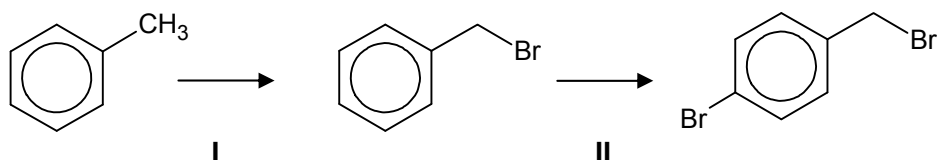
- A $\text{CCl}_2=\text{CHCH}_3$
 B $\text{C}_6\text{H}_5\text{CH}=\text{CFC}_6\text{H}_5$
 C $\text{C}_6\text{H}_5\text{CH}=\text{CH}(\text{CH}_2\text{Cl})$
 D $\text{CH}_3\text{CH}=\text{CH}(\text{CH}_3)$

- 17 What is the number of σ and π bonds present in a molecule of buta-1,3-diene?



	σ	π
A	3	2
B	6	2
C	9	2
D	11	0

18 Consider the following reaction sequence



What is the type of reaction for step I and II?

- | | I | II |
|----------|----------------------------|----------------------------|
| A | electrophilic substitution | electrophilic addition |
| B | electrophilic substitution | free radical substitution |
| C | free radical substitution | electrophilic addition |
| D | free radical substitution | electrophilic substitution |

19 Geraniol is found in rose oil. Upon controlled oxidation, geraniol yields CH_3COCH_3 , $\text{CH}_3\text{COCH}_2\text{CH}_2\text{COOH}$ and $\text{H}_2\text{C}_2\text{O}_4$.

What of the compounds listed below represents the structure of Geraniol?

- A** $\text{CH}_3\text{CH}=\text{C}(\text{CH}_3)\text{CH}_2\text{CH}_2\text{C}(\text{CH}_3)=\text{CHCH}_2\text{OH}$
- B** $(\text{CH}_3)_2\text{C}=\text{CHCH}_2\text{CH}_2\text{C}(\text{CH}_3)=\text{CHCH}_2\text{OH}$
- C** $(\text{CH}_3)_2\text{C}=\text{C}(\text{CH}_3)\text{CH}_2\text{C}(\text{CH}_3)=\text{CHCH}_2\text{OH}$
- D** $(\text{CH}_3)_2\text{C}=\text{CHCH}_2\text{CH}_2\text{C}(\text{CH}_2\text{OH})=\text{CHCH}_3$

20 When an organic compound **R** was treated with phosphorus pentachloride, fumes of hydrogen chloride were evolved. When **R** was warmed with acidified aqueous potassium dichromate(VI), the solution turned green.

Which of the following compounds could be **R**?

- A** $\text{CH}_3\text{CH}(\text{OH})\text{CH}_3$
- B** $\text{CH}_3\text{CH}_2\text{COOH}$
- C** $\text{CH}_3\text{CH}_2\text{CHO}$
- D** CH_3COCH_3

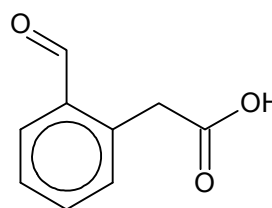
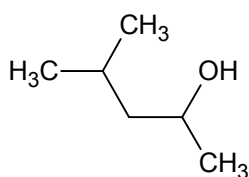
- 21 Hot ethanolic KOH is added to $\text{C}_6\text{H}_5\text{CH}_2\text{CH}_2\text{Cl}$ and $\text{C}_6\text{H}_5\text{CH}_2\text{Cl}$ separately. Predict if a reaction takes place and identify the possible product for each reaction if any.

	$\text{C}_6\text{H}_5\text{CH}_2\text{CH}_2\text{Cl}$ + Hot ethanolic KOH	$\text{C}_6\text{H}_5\text{CH}_2\text{Cl}$ + Hot ethanolic KOH
A	$\text{C}_6\text{H}_5\text{CH}=\text{CH}_2$	No reaction
B	$\text{C}_6\text{H}_5\text{CH}_2\text{CH}_2\text{OH}$	$\text{C}_6\text{H}_5\text{CH}_2\text{OH}$
C	$\text{C}_6\text{H}_5\text{CH}_2\text{CH}_2\text{OH}$	No reaction
D	$\text{C}_6\text{H}_5\text{CH}=\text{CH}_2$	$\text{C}_6\text{H}_5\text{CH}_2\text{OH}$

- 22 Which of the following reagents is most suitable for converting $\text{C}_6\text{H}_5\text{CH}_2\text{OH}$ to $\text{C}_6\text{H}_5\text{CH}_2\text{Br}$?

- A Br_2 (aq)
- B Br_2 with ultraviolet light and heat
- C Br_2 with FeBr_3 catalyst
- D PBr_3

- 23 Which of the following reagents may be used to distinguish between the compounds S and T?



- A Fehling's reagent.
- B Sodium metal.
- C Alkaline aqueous iodine.
- D Refluxed with potassium manganate (VII) in dilute H_2SO_4 .

- 24 Consider the 4 compounds below:

CH_3COOH	CHCl_2COOH	CH_2ClCOOH	$\text{CH}_3\text{CH}_2\text{OH}$
I	II	III	IV

Which of the following sequence arrange the compounds in order of increasing acid strength?

- A IV, I, II, III
 B IV, I, III, II
 C I, III, II, IV
 D IV, II, III, I
- 25 Which of the following gives the correct structural formula of the organic product of the reaction between $\text{CH}_3\text{CH}(\text{OH})\text{CH}_3$ and $\text{CH}_3\text{CO}_2\text{H}$?

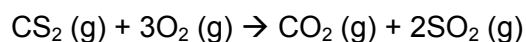
- A $\text{CH}_3\text{CO}_2\text{CH}(\text{CH}_3)_2$
 B $\text{CH}_3\text{CO}_2\text{CH}_2\text{CH}_2\text{CH}_3$
 C $(\text{CH}_3)_2\text{CHCO}_2\text{CH}_3$
 D $\text{CH}_3\text{OCOCH}_2\text{CH}_2\text{CH}_3$

For questions 26 – 30, the responses **A** to **D** should be selected on the basis of

A	B	C	D
1,2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is to be used as a correct response.

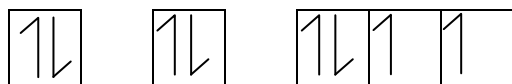
- 26** Carbon disulphide, CS₂, is a volatile flammable liquid used in the manufacture of cellophane. On combustion, CS₂ is oxidised as follows.



Which of the following statement(s) may be reflected from the above equation?

- 1** If 20 cm³ of CS₂ is reacted with 100 cm³ of oxygen, 60% of the final volume of gas will dissolve in alkali
- 2** 38.1 g of CS₂ will react exactly with 33.6 dm³ of O₂ under standard temperature and pressure.
- B 3** When 30 cm³ of CS₂ was reacted with 60 cm³ of oxygen, 30 cm³ of CO₂ will be produced.

- 27** The electronic configuration of species **Z** is shown below:



What of the following specie(s) could be **Z**?

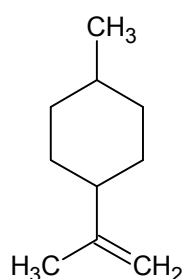
- 1** O
- B 2** Ne²⁺
- 3** F⁻

- 28 BeF_2 reacts readily with $(\text{CH}_3)_3\text{N}$ while NF_3 does not.

Which of the following statement(s) is/are correct?

- 1 The mole ratio of the amount of BeF_2 to the amount of $(\text{CH}_3)_3\text{N}$ required is 1:1
- 2 The shape of the resulting product is tetrahedral
- C** 3 There is an absence of available empty orbitals which prevented NF_3 from reacting with $(\text{CH}_3)_3\text{N}$

- 29 Limonene is the essential oil found in the skin of lemon and orange. Its structure is shown below.



Which statement(s) about the compound above is/are correct?

- 1 It can be oxidised by hot acidified KMnO_4 (aq).
- 2 It has *cis-trans* isomers.
- D** 3 It is a planar molecule.

- 30 A carbonyl compound, **X**, was reacted with 2,4-dinitrophenylhydrazine, and a pale yellow precipitate was observed. **X** also forms a pale yellow precipitate with alkaline aqueous iodine. **X** could be

- 1 $\text{CH}_3\text{COCH}_2\text{CH}_3$
- 2 $\text{CH}_3\text{CH}_2\text{CHO}$
- 3** $\text{C}_6\text{H}_5\text{COOH}$

END OF PAPER